

ACME Persistent DNS Challenge

draft-ietf-acme-dns-persist-01

Shiloh Heurich (Fastly)

Henry Birge-Lee (Crosslayer Labs)

Michael Slaughter (Amazon Trust Services)

IETF 125 ACME WG — Shenzhen

Since IETF 124 (Montreal)

Merged

- **PR #35 — accounturi decoupled from challenge object** — DNS record is self-contained; enables pre-provisioning, JIT validation, many-to-one URI model
- **PR #36 — Case-handling rules** explicit for all parameters
- **PR #41 — Renamed** "Authorization Domain Name" → "Validation Domain Name" (BR term conflict)

Implementation & Status

- **Pebble** testing CA — dns-persist-01 [merged](#); **lego** client — [merged](#)
- **cert-manager** — [feature request](#) open, [WIP implementation](#)
- -01 publication pending editorial fixes; five open issues, all on today's agenda

Subdomain Validation Gap (Issue #33)

The problem

Client has `_validation-persist` at `example.com` with `policy=wildcard`

Wants a cert for `server.dept.example.com`

No mechanism for the client to specify the parent domain

Without guidance, the CA either:

- Queries each ancestor (*expensive, ambiguous, MPIC amplification*)
- Checks only the FQDN → **fails**

DNS lookup today

```
example.com
└─ _validation-persist ← record
    policy=wildcard

dept.example.com ← no record
|
server.dept.example.com ← cert request
                        CA looks here
                        → nothing found
```

Proposed: `adn` Response Field

RFC 8555 §7.5.1 extension point — "future specifications might define [response fields]"

```
POST /chall/PAniVnsZcis/dns-persist-01
payload: { "adn": "example.com" }
```

- CA confirms `example.com` is a proper suffix of `server.dept.example.com`
- CA looks up `_validation-persist.example.com` → finds `policy=wildcard` → validates

Open questions:

1. **Depth limits:** Restrict label pruning? (*BRs allow pruning to base domain*)
2. **In this draft or separate?** (*We prefer this draft — same mechanism, ships in -02*)

TTL as Validation Ceiling (Issue #42)

Ceiling	Controlled by	Proposal
CA Validation Reuse Period	BRs / root programs	✓ Keep
<code>persistUntil</code> timestamp	Domain owner (explicit)	✓ Keep
DNS TXT Record TTL	DNS cache config	✗ Remove

Why remove?

- **Layer confusion** — TTL governs DNS caching, not CA policy; resolvers return *remaining* cache time, not authoritative TTL
- **Already bypassed** — Let's Encrypt caps resolved TTL to 1 min (*Ilari Liusvaara, Oct 2025*)
- **Operational breakage** — default TTLs of 30–60s would block issuance unintentionally

Question: Does any implementation use TTL to force revalidation? (*Only 2 have weighed in — seeking input*)

IP Validation via Reverse DNS (Issue #32)

Context

- CA/BF ballot SC-091 proposes dns-persist for IP addresses via `in-addr.arpa` / `ip6.arpa`
- Uses `_ip-validation-persist` label (distinct from `_validation-persist`)
- Same mechanism and parameters — different validation target and IANA registration
- **Google Trust Services** supports same-draft approach (*Frolov, Dec 2025*)

Question: Same draft or companion document?

Option	Pro	Con
Same draft	Less overhead, same mechanism	Depends on CA/BF ballot timeline
Companion	Independent timeline, cleaner IANA	More documents to track

Quick Items

Error Types (Issue #39)

- Three failure modes all return generic `unauthorized` — clients cannot distinguish
- No ACME extension has registered error types yet — **precedent question**
- **Suggestion:** Design the error space after #33 settles (new failure modes expected)
- *Is this blocking other work?*

Client-Side `persistUntil` Check (Issue #38)

- Should clients check `persistUntil` before responding to a challenge?
- **Concern:** Split-horizon DNS — enterprise clients may not see the record
- *Useful SHOULD guidance, or more confusing than helpful?*

Path Forward

Priorities for -02

1. Resolve TTL question — remove or keep ceiling (#42)
2. Spec `adn` field for subdomain validation (#33)
3. Decide IP validation scope — same draft or companion (#32)
4. Editorial cleanup from review findings

Timeline

- **-01:** Publish after editorial fixes (*imminent*) → **-02:** After subdomain + TTL resolution
- **WGLC target:** After -02 or -03

Implementation

- **Pebble** testing CA — merged; **lego** client — merged
- **Let's Encrypt** — **implementing**: staging late Q1, production Q2 2026
- **Fastly/Certainly** — committed for 2026; **Amazon Trust Services** — assessing

Questions & Discussion

Thank you!

Contact:

- Mailing list: acme@ietf.org
- GitHub: <https://github.com/ietf-wg-acme/draft-ietf-acme-dns-persist>
- Draft: <https://datatracker.ietf.org/doc/draft-ietf-acme-dns-persist/>